3 Week 3: Formal & Informal Technical Reports

3.1 Documenting the Introduction Chapter: Structuring the Project Problem Background Using the Inverted Pyramid Model

The introduction chapter of a research project or thesis sets the stage for the entire study and must be carefully constructed to effectively guide the reader from general context to the specific problem addressed.
The **Inverted Pyramid Model** is a useful structural metaphor for this purpose, especially in technical and computing research contexts.

. This model organizes information from the broadest context at the top, then narrows down progressively to the specific research problem. Starting with general background information, it funnels toward detailed problem statements and research questions, similar to how a news article delivers the most critical information upfront and then adds layers of detail Zhang and Liu (2016).

. **Applying the Model in Project Introductions:** For an IT project, the introduction typically begins by describing the broader technical domain or challenge, establishing the **research baseline**—the current state of knowledge, prior work, and technologies relevant to the problem. This creates a **contextual background** that grounds the study within the computing field.

. The next step narrows focus to highlight deficiencies, gaps, or specific issues in existing technologies, processes, or knowledge. This justifies the need for the computing solution you propose. This research gap defines the significance and originality of the project problem.

. Finally, the introduction culminates in a clear problem statement and the aims or objectives of the project, outlining what the research intends to accomplish or improve. This structured narrowing not only helps readers understand the rationale for the study but also demonstrates scholarly rigor in situating the project within the technical landscape Diyana Binti Maznun et al. (2017), Veerappan et al. (2024)

IT Example: Cloud Computing Security Project

1. **Broad Context (Top of the Pyramid):** Start by describing the general importance of cloud computing in modern IT infrastructure, emphasizing its widespread adoption for scalability and cost-effectiveness.

2. **Narrowing Context (Middle Layers):** Discuss common security challenges in cloud environments, such as data breaches, unauthorized access, and compliance issues. Present current security technologies and frameworks—like encryption methods, access control models, and intrusion detection systems—that attempt to address these challenges.

3. **Research Gap and Specific Problem (Bottom Layer):** Identify limitations in existing solutions—for instance, that most access control models don't adequately address dynamic user roles in multi-tenant cloud architectures. This gap justifies exploring a novel adaptive access control mechanism. Clearly state this as the problem your project intends to solve.

4. **Project Aims:** Conclude this section with a concise statement of objectives, such as designing and implementing an adaptive access control model that improves security responsiveness without compromising system performance.

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3.2 Establishing a Technical Research Baseline and Justifying the Required Computing Solution

- **Technical Research Baseline:** Setting up a research baseline involves reviewing prior work, existing technologies, and theoretical frameworks relevant to your project. In IT, this means summarizing relevant algorithms, systems, protocols, or software—essentially documenting what is known, proven, or currently practiced.
  By doing this, you demonstrate a **deep understanding** of the domain and avoid duplicating past work. You also identify opportunities where existing solutions fall short, creating space for your contribution.
  For example, in computational imaging or software development projects, presenting a baseline might involve discussing traditional algorithms or frameworks before introducing your novel approach Veerappan et al. (2024).

- **Justification of Computing Solution:** After establishing the baseline, you justify why a new or modified computing solution is required. This involves articulating the **limitations** or challenges in existing solutions—such as inefficiency, scalability issues, lack of accuracy, or user unfriendliness—and how your project is designed to overcome these issues.
  The justification also includes a discussion of the **potential impact or benefits** of the solution, such as improved performance, reduced costs, or novel capabilities.
  Incorporating citation of relevant literature and technical references strengthens the justification and situates your project within ongoing scholarly conversation, validating the research question's worthiness Diyana Binti Maznun et al. (2017).

IT Example: Machine Learning Model for Fraud Detection

1. **Technical Baseline:** Review existing fraud detection methods in financial IT systems, such as rule-based filters, logistic regression, and classical machine learning algorithms like decision trees and SVMs. Discuss their accuracy rates, scalability, and limitations like inability to adapt quickly to new fraud patterns.

2. **Justification for New Solution:** Justify the need for a new solution by stating issues with existing methods—perhaps they struggle with high false-positive rates or cannot process large volumes of streaming transaction data in real time.

3.3 Sub-Themes & Application: Contextual Framing, Technical Problem Statement Formatting, Introductory Citation Patterns

* Contextual framing in introductions means providing background information that positions the research problem within the larger technical and academic domain.
  For IT bachelor projects, this involves detailing the technological environment, key concepts, and preceding studies or experiments that relate to your problem area Veerappan et al. (2024).
  This framing ensures your audience understands the significance and relevance of the work, and how it connects to broader technical trends or challenges.
  Effective contextual framing also includes identifying existing knowledge gaps or unresolved issues in the field, which leads logically to your problem statement (Diyana Binti Maznun et al., 2017).

* **Technical Problem Statement Formatting Example for a software development project:** "While automated testing frameworks have been effective in many scenarios, current tools often lack support for asynchronous event-driven architectures, resulting in incomplete test coverage and delayed defect detection. This project aims to design a novel testing framework optimized for event-driven microservices architectures to enhance defect detection rates and reduce testing

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time".

Technical Problem Statement Formatting

- The problem statement should be clear, concise, and specific, explicitly articulating the core issue your study addresses.
- Establishing a topic area (general context).
- Identifying a gap, limitation, or problem in current knowledge or technology (research gap).
- Specifying the problem your research will tackle.
- Concluding with the purpose or objectives of the research.
- This structured approach aligns with models like Swales' Create-A-Research-Space (CARS) framework, highlighting the importance of "move 2" — counter-claiming or establishing the research gap, which is frequently challenging for students to articulate well Diyana Binti Maznun et al. (2017).
- The problem statement must be technically oriented and justified based on prior work presented in the baseline, making clear why the problem matters and what impact solving it could have.

* **Introductory Citation Patterns**: Proper citation in the introduction serves multiple functions: situating the project, showing familiarity with related work, and supporting claims about existing gaps or solutions.
Citation patterns generally include:

+ Referencing key foundational studies or techniques to ground your background.
+ Citing recent or highly relevant work that your project builds upon or differs from.
+ Using citations to validate the existence of the problem or limitations in current solutions.
+ Integrating citations smoothly within sentences, rather than a list of references, improves readability and scholarly tone Blicblau et al. (2015), Diyana Binti Maznun et al. (2017).

Example: E.g., "Recent research in global software engineering highlights increasing challenges in coordinating distributed teams due to cultural and language differences (Prikladnicki et al., 2013). However, current development process models inadequately address these complexities, motivating our project" (Alqahtani et al., 2024).

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Inverted Pyramid Model Application	Writing introductory chapter on optimizing database query performance by first explaining database systems generally, then tuning challenges, finally the specific indexing method improvement studied.
Technical Baseline Establishment	Summarizing existing cybersecurity threat detection frameworks before introducing a new ML-based anomaly detector.
Justifying Computing Solution	Explaining shortcomings in blockchain scalability that your project addresses with a novel consensus algorithm.
Citation Patterns	Citing foundational works on agile software development Rivera Ibarra et al. (2024), then recent improvements for motivation and productivity through gamification Meißner et al. (2025).
Contextual Framing	Framing your project within the IoT security domain by discussing the rapid expansion of IoT and its vulnerabilities Prikladnicki et al. (2013).
Problem Statement Formatting	Clearly identifying the gap: "Existing image classification algorithms fail to efficiently process multi-modal inputs in real-time applications" Chen and Blostein (2006).

Table 1: Additional Examples to Illustrate Key Concepts

3.4 Statement Hook

A **"statement hook"** is an opening sentence or phrase designed to immediately capture the reader's attention at the beginning of an IT project, thesis, or dissertation. It functions as a compelling entry point that introduces the topic in an engaging way, often highlighting the importance, relevance, or urgency of the research problem. In academic writing, especially in theses and dissertations, the statement hook sets the tone and motivates the reader to continue exploring the study Lewis et al. (2021), Prasetyanti and Tongpoon-Patanasorn (2023).

In an IT project or dissertation, a statement hook might highlight a pressing problem or an intriguing fact about the digital world to compel readers' interest.

For example:

Statement Hook: "Despite the rapid growth of cloud computing, over 60% of small businesses still struggle with secure and efficient data migration, leading to significant operational risks."

This hook immediately presents a striking statistic and a significant challenge, framing the problem's real-world impact and relevance. It entices the reader to learn more about the study's approach to resolving this issue.

Example in an IT Thesis Introduction In a thesis focused on improving cloud data migration processes, the introduction might start with:

"Cloud computing has revolutionized data storage and accessibility; however, organizations face persistent challenges in ensuring secure, low-latency migration of sensitive data to the cloud. This research addresses these challenges by developing adaptive migration algorithms tailored for heterogeneous IT environments."

Summary

** A statement hook is an attention-grabbing opening in an IT project, thesis, or dissertation introduction.

** It frames the problem or research context in a striking or relevant way.

** Effective hooks encourage readers to engage deeply with the research.

** **For example,** an IT project might hook readers by pointing out a widespread security risk or a technology gap in cloud computing or software development.

** Research on dissertation writing emphasizes the challenge and necessity of crafting such engaging introductions for scholarly success Lewis et al. (2021), Prasetyanti and Tongpoon-Patanasorn (2023).

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A Complete Introduction Example: "Improving Network Security in Small Businesses Using an Adaptive Firewall"

(Statement Hook) "In today's digital age, over 70% of small businesses face significant network security threats daily, yet many lack the technical resources to effectively defend their networks." This striking fact establishes the importance and urgency of addressing network security in small business environments.

(Background) Network security is a critical concern for organizations of all sizes, protecting sensitive data from cyberattacks and unauthorized access. Small businesses increasingly rely on digital networks for operations but often struggle with limited budgets and IT expertise. Existing security tools, including firewalls, are typically designed for larger enterprises, leaving small businesses vulnerable to evolving threats. Common firewall implementations in small enterprises tend to be static and inflexible, struggling to adapt to dynamic threats in real time.

(Technical Research Baseline) Conventional firewall technologies apply fixed rules to control network traffic, which can be insufficient for identifying new or sophisticated attack vectors. Recent studies have proposed adaptive firewall systems that dynamically adjust rules based on network behaviour and threat patterns. However, these adaptive solutions often require complex management and high computing resources, making them unsuitable for small business environments. Current commercial options lack scalability, user-friendliness, and affordable customization to the needs of small networks.

(Research Gap) While adaptive firewalls bring promise, there is a lack of practical solutions tailored specifically for small businesses with limited IT staff and budgets. Most solutions are either too complex or expensive for this market segment.

(Problem Statement) This project addresses the need to develop an adaptive firewall model designed for small business networks that balances effective dynamic threat mitigation with ease of deployment and management.

(Aim) The aim of this project is to design and implement a lightweight, adaptive firewall system that improves small business network security without requiring specialized IT expertise. **(Objectives)**

1. To analyse typical network threats faced by small businesses and existing firewall limitations.
2. To develop an adaptive firewall prototype based on simplified rule adjustment and anomaly detection.
3. To evaluate the prototype's effectiveness in detecting and mitigating common attacks under limited computational resources.
4. To ensure the system offers a user-friendly interface for non-expert administrators.

(Justification) Small businesses are increasingly targeted by cyberattacks but often rely on manual or static defences that cannot keep pace with threats. Deploying an adaptive firewall system specifically designed for small business constraints can significantly reduce security risks, operational disruptions, and potential financial losses. By focusing on lightweight algorithms and administrative simplicity, this project addresses the practical barriers preventing adoption of adaptive security solutions in the small business sector.

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3.5 The Grammar of Introduction.

Introduction Move	Recommended Tense	Underlying English Grammar Structure	Student-Facing Example
Broad Context & Universal Facts	**Present Simple**	**Subject + Verb 1 (s/es)** *(Used for active, undisputed computing facts and current technological realities.)*	"Cloud computing **[Subject] offers [Verb 1]** unprecedented scalability, and machine learning **[Subject] requires [Verb 1]** massive datasets."
Research Gap & System Vulnerabilities	**Present Simple**	**Subject + Verb 1 (s/es)** *(Used to articulate ongoing technical flaws, active threats, and system limitations.)*	"Static firewalls **[Subject] lack [Verb 1]** the flexibility to mitigate threats, and existing classifiers **[Subject] fail [Verb 1]** to process multi-modal inputs."
Technical Baseline (Historical Iterations)	**Past Simple (Active)**	**Subject + Verb 2 (Past Tense form)** *(Used to describe legacy software versions, obsolete protocols, or outdated systems.)*	"Early database frameworks **[Subject] utilized [Verb 2]** strict single-node architectures, while first-generation IoT devices **[Subject] relied [Verb 2]** on cleartext protocols."
Technical Baseline (Past Failures & Events)	**Past Simple (Passive)**	**Subject + was/were + Verb 3 (Past Participle)** *(Used when citing specific, documented historical security breaches or past system crashes.)*	"The regional energy distribution network **[Subject] was disrupted [was + Verb 3]** by a major cyberattack, and system memory **[Subject] was exhausted [was + Verb 3]** during early tests."
Project Aim & Objectives	**Present Simple**	**Subject + Verb 1 (s/es)** *(Used at the bottom of the pyramid to state what the current research intends to resolve.)*	"This project **[Subject] addresses [Verb 1]** the need for security responsiveness, and the prototype **[Subject] balances [Verb 1]** mitigation with ease of deployment."

Table 2: Split-Tense Protocol in Introduction Chapter

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References

Blicblau, A., Bruwer, M., and Dini, K. (2015). Do engineering students perceive that different learning and teaching modes improve their referencing and citation skills? *International Journal of Mechanical Engineering Education*, 44(1):3–15.

Chen, N. and Blostein, D. (2006). A survey of document image classification: problem statement, classifier architecture and performance evaluation. *International Journal of Document Analysis and Recognition (IJDAR)*, 10(1):1–16.

Diyana Binti Maznun, M., Monsefi, R., and Nimehchisalem, V. (2017). Undergraduate esl students' difficulties in writing the introduction for research reports. *Advances in Language and Literary Studies*, 8(1):9.

Lewis, K. B., Graham, I. D., Boland, L., and Stacey, D. (2021). Writing a compelling integrated discussion: a guide for integrated discussions in article-based theses and dissertations. *International Journal of Nursing Education Scholarship*, 18(1).

Meißner, N., Bredl, P., Speth, S., and Becker, S. (2025). Enhancing motivation in software engineering education through gamified agile project-based learning. In *Proceedings of the Special Interest Group on Computer Science Education (SIGCSE)*, pages 835–846.

Prasetyanti, D. C. and Tongpoon-Patanasorn, A. (2023). A move analysis of dissertation introductions written by native english speakers and indonesian phd students across disciplines. *rEFLections*, 30(2):468–487.

Prikladnicki, R., Boden, A., Avram, G., De Souza, C. R. B., and Wulf, V. (2013). Data collection in global software engineering research: learning from past experience. *Empirical Software Engineering*, 19(4).

Rivera Ibarra, J. G., Borrego, G., and Palacio, R. R. (2024). Early estimation in agile software development projects: A systematic mapping study. *Informatics*, 11(4):81.

Veerappan, V., Ahmad, M., Balakrishnan, K., Aris, M. A., and Suan, W. H. (2024). Genre analysis of introduction section in electrical engineering undergraduate laboratory reports. *F1000Research*, 11:163.

Zhang, H. and Liu, H. (2016). Visualizing structural "inverted pyramids" in english news discourse across levels. *Text & Talk*, 36(1):89–110.

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